

Parallel Machines Scheduling Problem for Minimization of Maximum Lateness with Sequence-Dependent Setup Times

Nasrin Mohabbati-Kalejahi

**Department of Industrial and Systems Engineering, Auburn University
Auburn, AL.**

Sang Won Yoon

**Department of Systems Science and Industrial Engineering, State University of New York at
Binghamton
Binghamton, NY.**

Abstract

In this paper, the identical parallel machines scheduling problem with sequence-dependent setup times is studied. The objective is to minimize the maximum lateness of jobs. A new mixed integer mathematical programming model is developed for solving the problem optimally. Since the research problem is shown to be NP-complete, two metaheuristic algorithms based on Genetic Algorithm (GA) and Simulated Annealing (SA) are applied to solve the problem. Random test problems are generated in different sizes to evaluate the efficiency of the proposed metaheuristic algorithms. The optimal solutions by MIP model are reported for the small size test problems. Computational results show that the proposed GA and SA algorithms can reach the optimal solutions for all of the proposed small size instances. For larger size test problems, the only an upper bound can be achieved by the MIP model. For these instances the proposed metaheuristic algorithms are converged to equal or better solutions than the upper bound of the MIP model in a significantly shorter time.

Keywords

Identical Parallel Machines Scheduling Problem, Mathematical Programming, Genetic Algorithm, Simulated Annealing.

1. Introduction

Rapid growth of today's competitive marketplace enforces manufacturers to provide a comprehensive coverage of customers' demand and reduce the production costs with efficient utilization of scarce resources. Scheduling is a substantial task, which has a significant role in decreasing production costs as well as improving productivity. Sometimes, producers are confronted with difficult scheduling problems such as parallel machines scheduling problems (PMSP), where there are multiple jobs that should be processed on multiple machines in parallel to optimize some measures of effectiveness.

A three part notation is introduced for describing the scheduling problems by [1] as $\alpha | \beta | \gamma$. In this notation, α , β and γ represent machine environment, processing characteristics and objective function of the scheduling problem, respectively. In this paper, jobs are assigned to be processed on m identical machines in parallel. Therefore, the machine environment notation in the proposed problem is represented as Pm . Processing characteristics of scheduling problems are important properties, which can contain various assumptions and make an intense impression on the complexity of the problem. One of the important processing characteristics in machine scheduling problem is the need of setup time when changing from one job to another in a sequence in order to prepare the machine, tool or process, which is called sequence-dependent setup time. Considering the setup time negligible or even ignorable adversely affects time-based performance measures. In order to perform a meaningful analysis on the completion time of jobs, a sequence-dependent setup time have been assumed in many practical situations such as paint operations, chemical industries, paper machines, automobile industry, and manufacturing shops [2]. Sequence-dependent setup time between each pair of jobs (j and k) is considered as the processing characteristics in this paper, which is represented as $s_{j,k}$.